3rdParty MCAL Integration

Release Notes

Infineon TC2xx

Version 1.0.2

Authors	Andrej Gazvoda
Status	Released

Document Information

History

Author	Date	Version	Remarks
Andrej Gazvoda	2017-03-22	1.0.0	Basic Integration of TC2xx MCAL Series
Andrej Gazvoda	2017-11-06	1.0.1	Added chapter 2.1.3.1
Andrej Gazvoda	2017-12-04	1.0.2	Update of chapter 2.1.3.1

Reference Documents

No.	Source	Title	Version
[1]	Vector	TechnicalReference_3rdParty-MCAL-Integration.pdf	See Delivery
[2]	Vector	ScreenCast_McalIntegration_Tresos.pdf	See Delivery

Scope of the Document

This document contains information about the integration of 3rd Party MCAL into Vector software stack.

Contents

1 MCAL Integration

1.1 Type of Integration

Basic Integration
Both configuration tools, EB tresos™ as well as DaVinci Configurator, are used for configuration.

Recommended workflow:
Start initial configuration with EB tresos™, export it in AUTOSAR format and import it into DaVinci Configurator. Generation and minor changes in configuration are done in DaVinci Configurator.
For usage with DaVinci Configurator 5 please refer to chapter 'Mixed configuration tool usage' in the document `TechnicalReference_3rdParty-MCAL-Integration.pdf` [1].

Furthermore, you will find a Multimedia Link in the document `ScreenCast_McalIntegration_Tresos.pdf` [2].

1.2 MCAL Location within SIP

The 3rd Party MCAL is separated from the Vector parts within the SIP. Furthermore, it might not be part of the delivery. Please refer to chapter 'First Steps' in document `TechnicalReference_3rdParty-MCAL-Integration.pdf` [1].

1.3 Supported 3rd Party Products

This integration supports the following NXP targets:

- TC22x, TC23x, TC26x, TC27x, TC29x

Note
Please refer to the Release Notes of the 3rd Party Products for further information, e.g. regarding supported versions, derivatives and compilers.

Note
Please be aware that only official 3rd Party vendor releases are part of this Vector integration package. Therefore any customer-specific releases cannot be considered.

Caution
Please contact the 3rdPartyVendor to find out if there are further Hotfixes available for your Mcal package.
It is essential to replace the effected module plugins in your original package before you start Script_MCAL_Prepare.bat.

1.4 Configuration Tools

1.4.1 TC2xx

- DaVinci Configurator 5.14 (and higher)

2 Vector Comment

Please consider the attached `TechnicalReference_3rdParty-MCAL-Integration.pdf` [1] for further information regarding Vector integration and setup of a project.

2.1 Known Issues

The original MCAL package might contains errors. Necessary patches can be provided via the SIP folder `ThirdParty\Mcal_TC2xx\VectorIntegration\Patches` (in the following named as 'Patch folder'.

2.1.1 All Derivatives

2.1.1.1 Creation of an EB tresos™ project

It is very important to configure the parameter `ResourceSubDerivative` in the component Resource *before* adding other components into the project. Otherwise, EB tresos™ will crash and cannot be closed anymore.

2.1.2 TC23x

2.1.2.1 Resource TC237 is not selectable

The module description files (BSWMD) for the TC237 derivatives are missing; this leads to the following consequences:

> TC237 derivatives cannot be selected via 3rd Party MCAL Integration Helper

> Vector tool chain cannot be used configuring and generating the dynamic MCAL sources as they are based on AUTOSRA module description files.

Workaround
Use EB tresos™ Studio for configuration and generation of the dynamic MCAL sources.

Infineon provided patches to allow selection of TC237 derivatives with the 3rd party helper tool and to use Vector tool DaVinci Configurator. Copy those files (if not received them from the MCAL vendor refer to the patch folder) into the 3rd party MCAL (SIP folder `\ThirdParty\Mcal_Tc23x\Supply\MC-ISAR_AS4XX_AURIX_TC23X_AB_PB_BASE_V400\TC23x_ABstep\Aurix_MC-ISAR\tools\tresosECU\bmd\AS4.0.3\TC23X`) keeping the folder structure. Then run the 3rd Party MCAL Integration Helper tool again.

2.1.3 TC22x

2.1.3.1 MC-ISAR_AS4XX_AURIX_TC22X_AC_PB_BASE_V200_REL_350 TC223L – Error in Mcu.bmd

There is a bug in Mcu.bmd for derivative TC223L in line 4236:

```
4235 <ECUC-ENUMERATION-LITERAL-DEF>
4236   <SHORT-NAME>TOUT42_SELB_PORT23_PIN1TOUT9_SELB_PORT00_PIN0</SHORT-NAME>
4237   <ORIGIN>INFINEON_AURIX</ORIGIN>
4238 </ECUC-ENUMERATION-LITERAL-DEF>
4239 <ECUC-ENUMERATION-LITERAL-DEF>
4240   <SHORT-NAME>TOUT8_SELB_PORT02_PIN8</SHORT-NAME>
4241   <ORIGIN>INFINEON_AURIX</ORIGIN>
4242 </ECUC-ENUMERATION-LITERAL-DEF>
```

The code sections must be patched as below:

```
4235 <ECUC-ENUMERATION-LITERAL-DEF>
4236   <SHORT-NAME>TOUT42_SELB_PORT23_PIN1</SHORT-NAME>
4237   <ORIGIN>INFINEON_AURIX</ORIGIN>
4238 </ECUC-ENUMERATION-LITERAL-DEF>
4239 <ECUC-ENUMERATION-LITERAL-DEF>
4240   <SHORT-NAME>TOUT9_SELB_PORT00_PIN0</SHORT-NAME>
4241   <ORIGIN>INFINEON_AURIX</ORIGIN>
4242 </ECUC-ENUMERATION-LITERAL-DEF>
4243 <ECUC-ENUMERATION-LITERAL-DEF>
4244   <SHORT-NAME>TOUT8_SELB_PORT02_PIN8</SHORT-NAME>
4245   <ORIGIN>INFINEON_AURIX</ORIGIN>
4246 </ECUC-ENUMERATION-LITERAL-DEF>
```

The issue has been reported to Infineon on 2017-11-06 and registered with https://jiraatvprod.intra.infineon.com/browse/0000051018-1709
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3 Glossary and Abbreviations

3.1 Glossary

Term	Description
3rd party components / MCAL	BSW modules not provided by Vector. Vector may have integrated the software within the SIP but does not take over any responsibility regarding functionality of these modules.
DaVinci Configurator	Configuration and generation tool for Vector MICROSAR components

Table 3-1 Glossary

3.2 Abbreviations

Abbreviation	Description
MCAL	Microcontroller Abstraction Layer
AUTOSAR	Automotive Open System Architecture
SIP	Software Integration Package (as provided by Vector)
Msn	Module Short Name according AUTOSAR

Table 3-2 Abbreviations

4 Contact

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